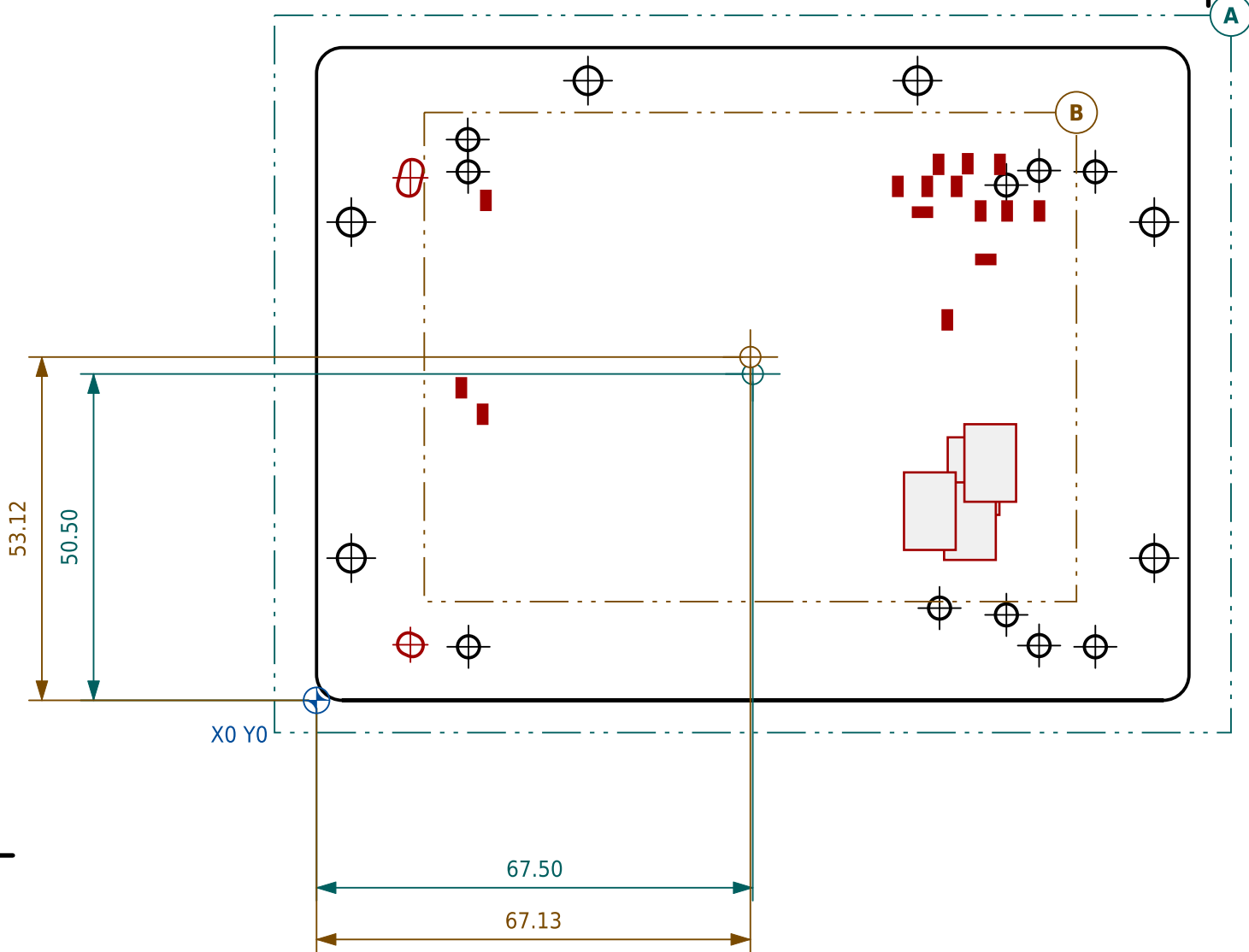




NOTES

1. The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
3. PLANE, which Z is measured from: A from the plate's own top face; B from the DEMO BOARD's top face, not the plate's -- the standoff height plus the board thickness, 1.56 to 1.60 mm, above the plate. The standoff height is the builder's and is specified nowhere here, so measure the stack. A stand set h mm too low covers only (Z - h) / Z of what is drawn, so the error always loses the edges.
4. HEADROOM: The demo board itself, any revision lies inside frame A in plan but stands above the plane its height is set from, and stays in the picture only up to 25.2 mm at 65 deg, 9.5 mm at 120 deg. Above that, raise the camera.

NOTES (continued)

5. FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
6. AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".
7. The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 44.26 mm in X, B by 30.18 mm in X -- which is still covered.
8. The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \text{atan}(\sqrt{3.76^2 + 2.74^2}) / 2 / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.

FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	Whole plate	148.00 x 111.00	X	67.50	50.50	146.8	55.5
B	Every LED and 7-seg	100.91 x 75.69	X	67.13	53.12	100.1	37.8

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

- Subject outline
- LED, display or connector
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- Dimension, extension and leader

NOTES (continued)

9. Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
10. Frame B is the union of every LED and 7-segment over all five revision families; no one revision needs all of it. They are not clustered -- 90.91 x 62.79 of a 135 x 101 plate -- so frame B buys little over frame A.

SOURCES

1. Plate geometry: [tinytapeout/mounting_plate/plate.py](#) - Outline and placements; see TT-MP-PLATE.
2. Indicator positions: [tinytapeout/boards.py](#) - LED and 7-segment footprints, in plate coordinates.
3. Raspberry Pi camera documentation: <https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
4. Arducam 5MP OV5647 documentation: <https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
5. Arducam motorized focus camera: <https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE TT MOUNTING PLATE

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-PLATE				REV A
SUBJECT TT Mounting Plate			GENERAL TOLERANCE plate +/-0.20, LEDs +/-0.10, Z DERIVED				